EXPERIMENT – 10

Perform Wireless Audit on Routers and Decrypt WPA Keys Using Aircrack-ng in Kali Linux

Aim

To perform a wireless security audit on Wi-Fi routers and demonstrate WPA/WPA2 key cracking using the Aircrack-ng suite in Kali Linux with an external USB Wi-Fi adapter.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Wireless Audit Tool:** Aircrack-ng
- **Hardware Requirement:** External USB Wi-Fi Adapter (Monitor Mode supported)
- **System Requirements:**
  - RAM: Minimum 4 GB (8 GB recommended)
  - Disk Space: Minimum 30 GB
  - Processor: Intel/AMD with Virtualization support

Introduction

Wireless networks are widely used but are vulnerable to attacks if weak encryption or passwords are used.
Aircrack-ng is a suite of tools used for monitoring, attacking, testing, and cracking Wi-Fi networks.
This experiment demonstrates how WPA/WPA2 handshakes can be captured and cracked to analyze the strength of wireless security.
Kali Linux with an external Wi-Fi adapter provides a controlled environment for ethical wireless security testing.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Connect the external USB Wi-Fi adapter and attach it to the VM.

Step 2: Verify Wireless Adapter

1. Open Terminal.
2. Check wireless interfaces:

3. iwconfig
4. Identify the wireless interface (e.g., wlan0).

Step 3: Enable Monitor Mode

1. Enable monitor mode:

2. sudo airmon-ng start wlan0
3. Monitor interface (e.g., wlan0mon) is created.

Step 4: Scan Nearby Wireless Networks

1. Scan for Wi-Fi networks:

2. sudo airodump-ng wlan0mon
3. Note the **BSSID**, **Channel**, and **Encryption type** of the target network.

Step 5: Capture WPA Handshake

1. Lock onto the target network:

2. sudo airodump-ng -c <channel> --bssid <BSSID> -w capture wlan0mon
3. Wait for a client to connect, or deauthenticate a client.

Step 6: Deauthenticate Client (Optional)

1. Force client reconnection:

2. sudo aireplay-ng --deauth 5 -a <BSSID> wlan0mon
3. WPA handshake is captured.

Step 7: Crack WPA Key

1. Use Aircrack-ng with a wordlist:

2. sudo aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt
3. Observe password cracking process.

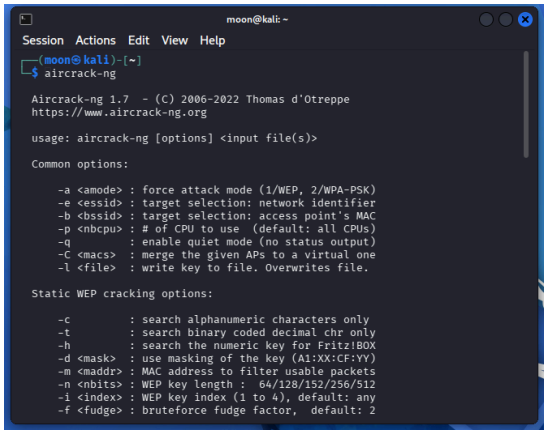
Step 8: Stop Monitor Mode

1. Stop monitor mode:

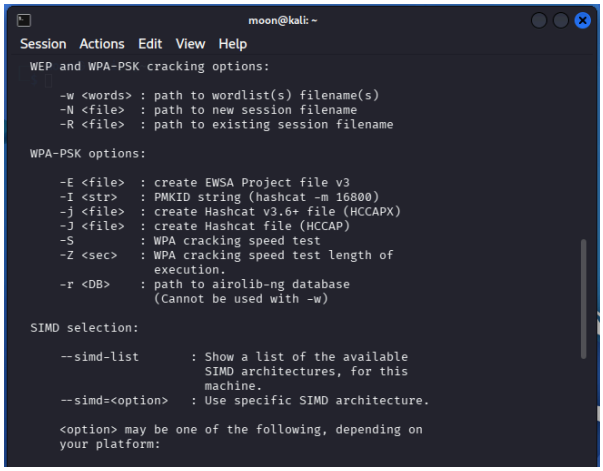
2. sudo airmon-ng stop wlan0mon

Output

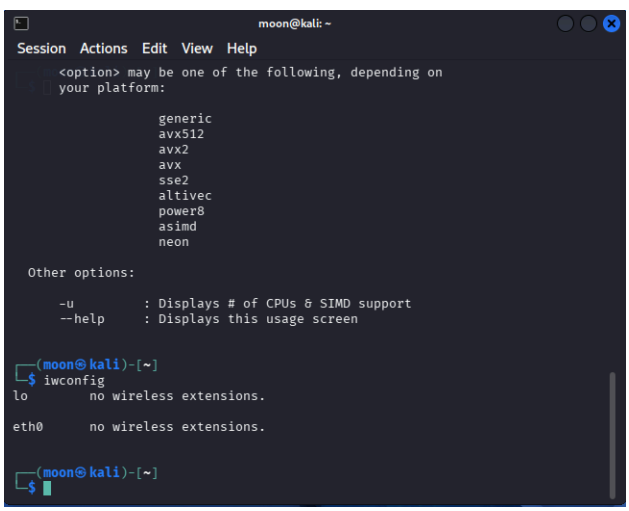
- WPA/WPA2 handshake successfully captured.
- Password cracking attempt performed.
- Weak passwords are recovered successfully.



```
moon@kali: ~
Session  Actions  Edit  View  Help
┌──(moon㉿kali)-[~]
└─$ aircrack-ng

  Aircrack-ng 1.7  - (C) 2006-2022 Thomas d'Otreppe
  https://www.aircrack-ng.org

  usage: aircrack-ng [options] <input file(s)>

  Common options:

      -a <amode> : force attack mode (1/WEP, 2/WPA-PSK)
      -e <essid> : target selection: network identifier
      -b <bssid> : target selection: access point's MAC
      -p <nbcpu> : # of CPU to use  (default: all CPUs)
      -q         : enable quiet mode (no status output)
      -C <macs>  : merge the given APs to a virtual one
      -l <file>  : write key to file. Overwrites file.

  Static WEP cracking options:

      -c         : search alphanumeric characters only
      -t         : search binary coded decimal chr only
      -h         : search the numeric key for Fritz!BOX
      -d <mask>  : use masking of the key (A1:XX:CF:YY)
      -m <maddr> : MAC address to filter usable packets
      -n <nbits> : WEP key length :  64/128/152/256/512
      -i <index> : WEP key index (1 to 4), default: any
      -f <fudge> : bruteforce fudge factor,  default: 2
```



```
moon@kali: ~
Session  Actions  Edit  View  Help
  WEP and WPA-PSK cracking options:

      -w <words> : path to wordlist(s) filename(s)
      -N <file>  : path to new session filename
      -R <file>  : path to existing session filename

  WPA-PSK options:

      -E <file>  : create EWSA Project file v3
      -I <str>   : PMKID string (hashcat -m 16800)
      -j <file>  : create Hashcat v3.6+ file (HCCAPX)
      -J <file>  : create Hashcat file (HCCAP)
      -S         : WPA cracking speed test
      -Z <sec>   : WPA cracking speed test length of
                   execution.
      -r <DB>    : path to airolib-ng database
                   (Cannot be used with -w)

  SIMD selection:

      --simd-list       : Show a list of the available
                          SIMD architectures, for this
                          machine.
      --simd=<option>   : Use specific SIMD architecture.

      <option> may be one of the following, depending on
      your platform:
```



```
moon@kali: ~
Session  Actions  Edit  View  Help
      <option> may be one of the following, depending on
      your platform:

                    generic
                    avx512
                    avx2
                    avx
                    sse2
                    altivec
                    power8
                    asimd
                    neon

  Other options:

      -u         : Displays # of CPUs & SIMD support
      --help     : Displays this usage screen


┌──(moon㉿kali)-[~]
└─$ iwconfig
lo        no wireless extensions.

eth0      no wireless extensions.


┌──(moon㉿kali)-[~]
└─$
```

Result

Wireless auditing and WPA key decryption were successfully demonstrated using the Aircrack-ng tool in Kali Linux.
The experiment highlights the importance of strong passwords and secure wireless encryption methods.